



Data-driven strategies for digital native market segmentation using clustering

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ARTICLE INFO

Keywords:

Digital native
Social networking sites
Market segmentation
Clustering
Digital business tools

ABSTRACT

The rapid growth of internet users and social networking sites presents significant challenges for entrepreneurs and marketers. Understanding the evolving behavioral and psychological patterns across consumer demographics is crucial for adapting business models effectively. Particularly, the emergence of new firms targeting adolescents and future generations underscores the importance of comprehending online consumer behavior and communication dynamics. To tackle these challenges, we introduce a Machine Learning-based Digital Native Market Segmentation designed to cater specifically to the interests of digital natives. Leveraging an open-access prototype dataset from social networking sites (SNS), our study employs a variety of clustering techniques, including Kmeans, MiniBatch Kmeans, AGNES, and Fuzzy C-means, to uncover hidden interests of teenage consumers from SNS data. Through rigorous evaluation of these clustering approaches by default parameters, we identify the optimal number of clusters and group consumers with similar tastes effectively. Our findings provide actionable insights into business impact and critical patterns driving future marketing growth. In our experiment, we systematically evaluate various clustering techniques, and notably, the Kmeans cluster outperforms others, demonstrating strong segmentation ability in the digital market. Specifically, it achieves silhouette scores of 63.90% and 58.06% for 2 and 3 clusters, respectively, highlighting its effectiveness in segmenting the digital market.

1. Introduction

Digital Native was introduced by Marc Prensky in 2001 to refer to a generation that has been raised in an era of ubiquitous technology, encompassing computers and the Internet (Marc, 2001). Digital natives feel comfortable with technology and computers at an early age and see technology as an integral and necessary part of their lives. Many adolescents and children in developed countries are considered digital natives because they primarily communicate and learn through computers, social networking services, and texting (Bottaro & Faraci, 2022). Digital natives are inspired by the new generation of social network services (SNS) and the Internet, and they create and share their moments to impress other people (Thimm, 2023). A social networking service (SNS) is an online platform that is used by people to create social networks or social relationships with others who share similar personal or professional content, interests, activities, backgrounds, or real relationships (Peters, Rohr, & Squires, 2022). Teenagers are completely

immersed in the digital world. Entrepreneurs should analyze their areas of interest for the benefit of their business. To this day, failing to establish a robust presence on social network services poses a significant risk to companies, potentially resulting in missed opportunities for growth and success (Mazzocchini & Lucarelli, 2023). A successful marketing strategy for social networking can increase an organization's brand awareness, increase customer satisfaction, and engagement, and encourage the organization to develop its voice (Cartwright & Davies, 2022).

Data mining algorithms, including clustering and classification, offer valuable assistance to businesses in uncovering insightful patterns within their customer data, computer network and security data (Talukder, Hasan, et al., 2023; Talukder et al., 2022; Talukder, Islam, et al., 2023, 2024; Talukder, Alamin, Layek, Kazi, Uddin & Aryal, 2023). These algorithms analyze demographic and behavioral

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<https://doi.org/10.1016/j.ijcce.2024.04.002>

Received 30 October 2023; Received in revised form 30 April 2024; Accepted 30 April 2024

Available online 7 May 2024

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information, enabling companies to tailor their marketing campaigns to customer interests effectively (Hasan et al., 2023; Wang, 2022). Machine learning and deep learning are widely used techniques in the modern world to solve real-world problems (Almomani et al., 2024; Balasubramani, Aravindhar, Renjith, & Ramesh, 2024; Kaur, Singh, & Kaushal, 2024; Krishnamoorthy, Sathiyarayanan, & Proença, 2024). Clustering algorithms, a type of unsupervised machine learning strategy, play a crucial role in grouping unbranded data based on similarities and disparities among features within a dataset (Demircan, Seller, Abut, & Akay, 2021; Ezugwu et al., 2022; Zhao et al., 2023). This approach facilitates the identification of meaningful clusters, enabling businesses to gain a deeper understanding of their data and make informed decisions (Bhattacharjya et al., 2024; Islam, Adil, et al., 2023; Islam, Talukder, et al., 2023; Khatun et al., 2023; Rana et al., 2023; Sharmin, Ahammad, Talukder, & Ghose, 2023; Talukder, Hossen, Uddin, Uddin & Acharjee, 2024; Talukder, Sharmin, Uddin, Islam & Aryal, 2024).

Many researchers (Nikou, Mezei, & Brännback, 2018; Smith, 2019; Umuhoza, Ntirushwamaboko, Awuah, & Birir, 2020) have applied different clustering approaches to analyze digital markets. For example, Nikou et al. (2018) investigated how pricing systems impact the engagement of digital natives with social media using data from 116 social media users. Smith (2019) examined the preferences of digital natives about mobile ads. Their study emphasized the importance of mobile ads in reaching the fast-growing population of smartphone users. Wong, Tan, Hew, Ooi, and Leong (2020) focused on mobile social media marketing and its implications for digital natives. The study provided valuable information on the factors that affect the population of mobile technology users, enhancing our understanding of mobile social media marketing. Umuhoza et al. (2020) proposed a behavior-based segmentation model for African credit cardholders.

Our research addresses the pressing need to understand and adapt to the evolving dynamics of digital markets, particularly in the context of the increasing influence of digital natives and the pervasive use of social networking services (SNS). By delving into the behavioral patterns of digital natives and their interactions with SNS platforms, we aim to provide valuable insights for marketers and entrepreneurs seeking to optimize their strategies and cater to this growing demographic effectively.

Furthermore, our study builds upon existing research in digital market analysis and customer segmentation, leveraging advanced clustering techniques and data mining algorithms to uncover meaningful patterns within SNS data. While previous studies have explored similar themes, our research contributes by offering a comprehensive analysis framework that integrates traditional and innovative clustering methods. This approach enables us to extract nuanced insights from SNS data and inform targeted marketing strategies tailored to the interests and preferences of digital natives.

The motivation behind our research stems from the imperative to stay ahead of the curve in an increasingly competitive digital landscape. By gaining a deeper understanding of digital market dynamics and consumer behavior, organizations can enhance their marketing effectiveness, improve customer engagement, and drive business growth. Ultimately, our aim is to empower businesses with actionable insights derived from sophisticated data analysis techniques, thereby enabling them to thrive in the digital age.

In addition to customer segmentation policies explored in various domains, our research contributes to analyzing customer market segments. Our contributions are summarized as follows:

1. Development of a Clustering-Based Market Analysis Model:

Our goal is to develop a model using advanced clustering techniques to analyze market data, focusing on extracting user interests that can enhance marketing sales. This model will blend traditional and innovative clustering methods to accurately identify and predict market trends.

2. **In-Depth Market Analysis Using Clustering Techniques:** We aim to apply a variety of clustering techniques for a thorough examination of market data, uncovering hidden patterns that provide critical insights for marketing strategists. This analysis is key to understanding the evolving dynamics of the digital market.

3. **Segmentation and Categorization of Customer Groups:** Our objective is to segment customers into distinct groups based on shared interests, using data categorization to inform targeted marketing strategies. This segmentation will also facilitate future research using supervised learning for deeper insights into customer behaviors and preferences.

The remainder of the paper gives a brief outline of our planned study, which includes Related works, Proposed Methodology, Experiment setup and Evaluation, and finally Conclusions and future work in Section 2, Section 3, Section 4 and Section 6 respectively.

2. Related works

The customer segment is the customer base business, which is divided into groups called customer segments so that each customer segment carries customers who share similar market features. These differences are based on factors that may directly or indirectly affect the market or activity, such as product preferences or expectations, location, behavior (Naim, 2023)

This section reviews some relevant work, from the literature on adolescent segregation, customer segregation, and the effects of SNS on adolescents. In this paper, we have focused mainly on behavioral and psychological customer segmentation. The key to this paper is to identify the segments of adolescent customers of the business using a clustering method.

2.0.1. Digital natives and mobile social network services

Manor and Kampf (2022) examined generational differences within foreign ministries and their impact on communication goals. It focused on digital nativity and identified operational and conceptual gaps between digital natives and immigrants. With a sample of 133 diplomats from six foreign ministries, the study found minimal operational gaps but significant conceptual gaps between the two generations. Digital immigrants tended to use social networking sites for one-way communication and were less engaged with follower feedback compared to digital natives. These gaps might have hindered effective policy marketing and formulation. The study offered policy recommendations to bridge these divides.

Choi, Jung, Kim, and Kim (2020) conducted a study on the competitive dynamics among prominent social networking platforms in the South Korean market, including Facebook, Instagram, and Twitter. The aim was to propose and test hypotheses regarding their competitive relationships and to analyze the trends within the domestic SNS market. Employing the Competitive Lotka–Volterra (LV) model, the study examined these relationships and utilized the Moving Window technique to illustrate their temporal changes. Through experimentation with the Moving Window technique, the research identified the interplay among the platforms and tracked their evolving dynamics over time.

Azzaoui, Singh, and Park (2021) proposed an SNS Big Data Analysis Framework for predicting COVID-19 outbreaks in Smart Sustainable Healthy Cities, utilizing Twitter data. Over two months, 10,000 Tweets were collected, revealing demographics such as 38% aged 18–29 and 56% male users. Natural Language Processing (NLP) filtered the tweets, enabling the prediction of outbreak clusters seven days before confirmed cases, with a high indicator of 0.989. This early detection potential aided in scientifically reducing infection rates in smart, sustainable, healthy city environments.

Andrews, Bianchi, Wiese, Cuneo, and Fazal E. Hasan (2019) examined consumer segmentation and responsiveness to marketing on social

network sites (SNSs) amid the growing consumer presence on these platforms. Conducting an online survey across Australia, Chile, South Africa, and Spain with 1025 respondents, the research identified four primary consumer segments: Social Seekers, Social Satellites, Social Fringe, and Disinterested, validated across all four countries. Additionally, a fifth segment, Social Tourists, emerged specifically in the South African data. The study provided valuable insights for marketers and practitioners seeking to target and engage consumers effectively on SNSs.

Nikou et al. (2018) discuss the intention of digital natives to interact with social media. They used data from 116 social media users and attempted to investigate how pricing systems affect digital natives' intention to interact with social media. The results of a configuration analysis show that gender seems to influence many configurations leading to interesting results.

Smartphones are the fastest-growing digital medium. Mobile ads occur on any mobile device like smartphones and tablets. The next generation of digital natives is especially prone to watching mobile ads. Digital Native has grown with Digital Communication. Digital migrants use their smartphones to connect with the world and shop. Smith (2019) Investigate the preferences of digital natives regarding mobile advertising. This paper provides some recommendations such as how mobile ads should respond within 5 s; Providing consumers with control over ad avoidance; and Advertising must provide simple, straightforward information that will benefit you.

Wong et al. (2020) provides valuable insights into mobile social applicability and digital native media marketing. Data collected from 263 university students were analyzed using structural equations based on partial minimal square modeling. This study determines the progress of knowledge about the emerging field of mobile social media marketing, which considers the factors affecting the population of mobile technology users.

2.0.2. Adolescent behavioral market segmentation

Umhuoza et al. (2020) define and describe the steps that can be taken to create a behavior-based segmentation model that differentiates African credit cardholders who make their purchases based on data. They used K-means clustering algorithms for customer segmentation and silhouette coefficients for performance evaluation. The results can help marketers make their campaigns customer-centric and reduce the associated marketing costs.

Disrupted communication and information technology have changed the political, social, technological and economic aspects of society. It is one of the biggest influences on the new generation, which differs completely from the previous ones. Generation Z will be the main consumer and viewer of the Internet. Therefore, it is important to identify their characteristics and behavioral patterns on popular platforms. Kolli, Salavatian, Khajeheian, and Yazdani said that big data needs to be noticed to identify the type of behavior. The sample selected a dataset of Instagram users born between 1995 and 2012. The results show that Generation Z is more interested in photo content than other content; They comment more than share pictures; Adolescents have a subculture, and speech around the central signal is visibly acceptable. Allaway et al. showed a 90 percent failure rate of 30,000 new products due to weak market segmentation (Allaway, McCullough, Kim, & McAmis, 2022).

Indartoyo (2021) state that, E-commerce players see the increase in Internet user penetration in Indonesia as a huge opportunity to expand their business. While this is beneficial for consumers, it has become a challenge for current and future markets. The segmentation in this study was done by grouping 455 respondents in Jakarta who have similar aspects in both population and behavior. The results of this study indicate the existence of three distinct divisions: neonatal buyer, general buyer and dedicated buyer, each with its own demographic and behavioral characteristics that are very interesting to study.

2.0.3. Adolescent psychological market segmentation

Ranto, Hariningsih, Prasetyanto, and Oktafiani (2021), The goal of this study is to examine the relationship between discount price and the relevance of fashion to buying passion. Self-governing questionnaires were created from the literature conducted by 135 female adolescents in Yogyakarta. This study provides valuable insights into teens' shopping behavior in an online setting, especially in the Shop Marketplace, where, as before, integrated price discounts and fashion are included.

2.0.4. Other customer segmentation policy

Qadadeh and Abdallah (2018), segmented insurance company datasets and used K-minus and self-organized maps. This paper provides the idea of big data analysis in customer segmentation.

Lorenzo-Romero, Constantinides, and Alarcón-del Amo (2012), discuss that SNS as a domain offers many opportunities to small and medium-sized enterprises, attracting growing online customers but understanding the online network customers is the basis for developing effective marketing programs. The research paper presents the results of a survey on SNS users in the Netherlands, discusses the impact of results management and identifies future research areas.

Due to the rapid growth of population and communication technology in the last decade, the use of social networks is increasing and is being used for various purposes. Hassan, Hussain, Hussain, Sadiq, and Lee (2017) examined emotions from text, using emotion theory, machine learning techniques, and natural language to examine a person's level of depression in social media posts.

3. Methodology

In this section, we discuss unsupervised machine learning approaches and show a detailed description of the knowledge mining architecture. The study finds patterns and structures of user's interests. The algorithms we used in this paper are Kmeans, Mini-Bach Kmeans, Hierarchical Agglomerative and Fuzzy C-means.

3.0.1. Proposed teenage clustering system

We present a clustering approach for the adolescent customer in Fig. 1. The model uses four clustering algorithms where the first three algorithms are known as hard clustering and the Fuzzy C-values are known as soft clustering algorithms. The performance of the cluster algorithm varies depending on the dataset's quality. We used Python as our programming language and Jupiter Notebook as our IDEA. The Workflow diagram of the proposed market segmentation method is shown in Fig. 2.

• Input Dataset:

We used the pandas library for our dataset management. Adolescents have different areas of interest in different sections of the dataset. These datasets can be collected from social network site surveys (SNS).

• Preprocessing:

There are some important steps that need to be performed before clustering datasets. Real-world data might not fit into the ML model. We apply some preprocessing approaches to improve the quality of the dataset. In this section, we have summarized the datasets; removed the null value, object variables converted to numerals; removed outliers and standardized the data.

• Apply Clustering algorithms:

Then we apply 4 clustering algorithms such as k-means, mini-batch k-means, hierarchical (AGNES) and fuzzy c-means clustering. For k-means and mini-batch k-means, we used the elbow method, for hierarchical we used dendrogram and for fuzzy c-means clustering we used PC and PE to find the optimal number of clusters.

• Clustering performance analysis:

Finally, to evaluate the performance we used silhouette coefficient and data clustering for each algorithm. Moreover, we analyzed the average interest score as well as the probability of interest of teenagers in each cluster.

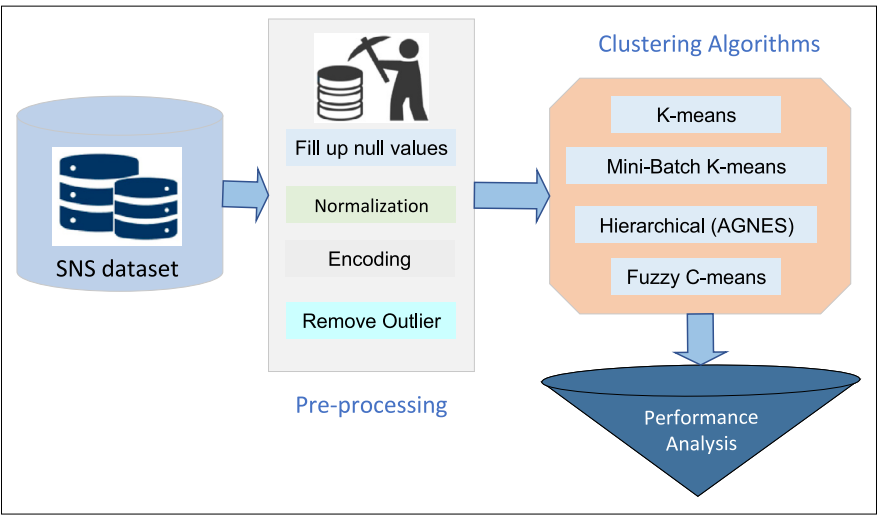


Fig. 1. Overall architecture of the proposed market segmentation architecture.

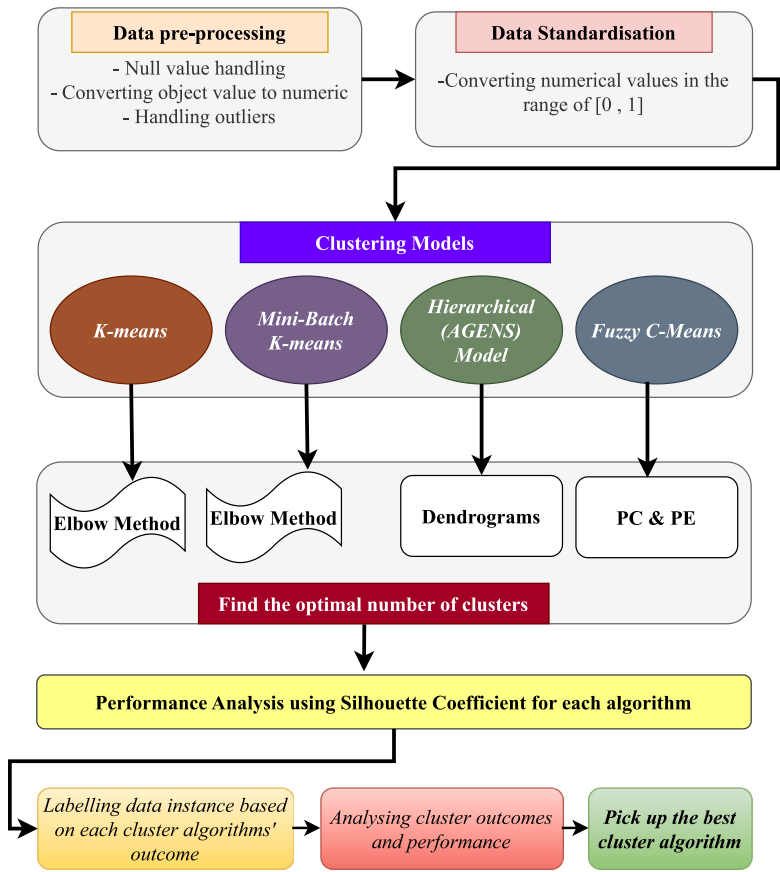


Fig. 2. Workflow diagram of the proposed market segmentation method.

3.1. Clustering algorithms

Clustering is the division process of a data point in multiple groups so that data points of the same group are more similar to other data points of the same group and different from the data points of other groups (Nazari, Hussain, & Musilek, 2023).

Hard clustering: Each data point is grouped into a cluster or any one of the clusters. It is binary-based. Example: K-means, K-Medoids.

Soft clustering: Each data point can belong to multiple clusters with its probability score or probability. It is probability-based. Example: Fuzzy C-means.

3.1.1. Kmeans clustering

For K-means, the elbow method is one of the most popular methods for selecting the optimal cluster number (Patel, Sivaiah, & Patel, 2022). We need to select the value of k in the elbow curve, where the distortion/inertia decreases linearly. K means K shows the number of

clusters and each cluster will have a centroid (Ashari, Nugroho, Baraku, Yanda, & Liwardana, 2023).

- Determine the number of clusters — elbow method.
- Select the K cluster center at random.
- Assign data points to the nearest K.
- Calculate the cluster center for each cluster, which is known as the centroid.
- Build clustering model for target cluster.
- Finally, extract the pattern from the clustering result.

K-means Clustering Equation:

$$S_k = \sum_{i=1}^n \min_{\mu_j \in C} (\|x_i - \mu_j\|^2) \quad (1)$$

Where S_k is the within-cluster sum of squares, x_i represents each data point, μ_j is the mean of cluster C , and n is the number of data points. The objective is to minimize the distance between data points and their respective cluster centroids.

3.1.2. Mini-batch K-means clustering

The Mini-Batch K-Means clustering algorithm is a version of the K-Means algorithm that can be used instead of the K-Means algorithm when clustering large datasets. When working on large datasets, it does not replicate the entire dataset (Ikotun, Ezugwu, Abualigah, Abuhaija, & Heming, 2023). This creates a random batch of data to store in memory, then randomly collects data at each iteration to update clusters. The mini-batch K-means algorithm reduces the computational cost of finding a cluster. Mini-batch K-means can give better results when we work on a larger dataset (Hanji & Hanji, 2023).

- In the first step, it has randomly drawn samples from the dataset to create a mini-batch. It then assigned these to the nearest centroid.
- In the second step, the centroid is updated. In contrast to k, this is done on a per-sample basis.
- For each sample of the mini-batch, the streaming average of the assigned centroid sample and all previous samples assigned to that centroid are updated.
- It has the effect of reducing the rate of change for a centroid.
- These steps are performed until combined or a pre-determined number of repetitions.

Mini-Batch K-means Clustering Equation: Similar to K-means, but the update step is modified for batch processing:

$$\mu_j = \mu_j + \eta(x' - \mu_j) \quad (2)$$

Where μ_j is the cluster centroid, x' is a sample from the mini-batch, and η is the learning rate. This equation adjusts the centroids based on a subset of data.

3.1.3. Hierarchical agglomerative clustering

In K-means, we need to determine the number of clusters at the beginning of the algorithm. Ideally, we did not know how many clusters we should have at the beginning of the algorithm and so this is a challenge with K-means. Classified clustering eliminates the problem of pre-defining the number of clusters. It can be created on a distance matrix or on a raw dataset for which the distance is incomplete. Start with a random choice of clusters so K-means that the output may change in algorithms where the output produced by the classified clustering does not change. Agglomerate clustering is a bottom-up clustering approach. We initially considered each observation as a cluster within itself and it formed a cluster by combining the nearest ones and moving towards the classification (Ezugwu et al., 2022). The method is simple and easy to explain.

- Cluster each observation into a single point.

- Pick two observations close to each other and combine them into one cluster.
- Choose the next two clusters that are close to each other and combine them into one cluster.
- Repeat only up to one cluster.

Hierarchical Agglomerative Clustering Equation: This technique is based on a linkage criterion, such as:

$$D(X, Y) = \min_{x \in X, y \in Y} d(x, y) \quad (3)$$

Where $D(X, Y)$ is the distance between clusters X and Y , and $d(x, y)$ is the distance between data points x and y . The method iteratively merges the closest clusters.

3.1.4. Fuzzy C-means clustering

Fuzzy C-means clustering is a soft clustering method where each data point is assigned a probability score within that cluster. A point of data could belong to several clusters (Rad et al., 2022). As data points overlap in most cases, it has more accuracy. It is more suitable for obscure clustering problems. The sum of each data point must equal one (Wang, Wang, & Wang, 2022). It is better suited for fuzzy clustering problems.

We can calculate the value of PC and PE to verify the fuzzy clustering. The partition coefficient (PC) measures the opacity level of the final split cluster and the larger the value, the better the partition result (Wongkhuenkaew, Auephanwiriyakul, Theera-Umpon, Teeyapan, & Yeesarapat, 2023). Partition entropy (PE) measures the opacity of the final split cluster, and the smaller the value, the better the partition results (de Almeida, Rizzo, Amorim, dos Santos, Rosas, Campos, Rosin, Zabini, & Demattê, 2023).

- Randomly select 'c' number of cluster centers.
- Calculate cluster membership probability score belonging to that cluster.
- Compute the fuzzy cluster centers again.
- Repeat the previous two steps until a minimal value is archived.

Fuzzy C-means Clustering Equation: The objective function for Fuzzy C-means is:

$$J_m = \sum_{i=1}^n \sum_{j=1}^c u_{ij}^m \|x_i - \mu_j\|^2 \quad (4)$$

Where u_{ij} is the degree of membership of x_i in the cluster j , μ_j is the centroid of cluster j , m is a fuzziness index, and n is the number of data points. The aim is to minimize J_m , which represents the total variance within the clusters.

3.2. Clustering performance evaluation

We used silhouette coefficients to measure clustering performance. The silhouette score is a metric used to measure the quality of a clustering algorithm. Its values range from -1 to 1 . A higher silhouette coefficient score corresponds to a model with a well-defined cluster. The silhouette coefficient (s) is defined for each sample and comprises two scores:

$$s = \frac{b - a}{\max(a, b)} \quad (5)$$

a: The average distance between a sample and all other points of the same class.

b: The average distance between a sample and all the other points in the next nearest cluster (Quraishi & Choudhury, 2023)

Table 1
Five categories to represent the 36 columns of interest.

Category	Grouped columns
Athletes	basketball, football, soccer, softball, volleyball, baseball, tennis, cheerleading, swimming, sports
Extracurricular activities	dance, band, music, rock, marching, cute, sexy, hot
Fashion style	hair, dress, shopping, clothes, hollister, abercrombie, mall, blonde
Religious education	god, jesus, bible, church
Antisocial behavior	sex, kissed, die, death, drunk, drugs

4. Experimental setup

In this section, we detail our experimental approach focused on segmenting adolescent Social Networking Services (SNS) users using various clustering algorithms. Our methodology involved data preprocessing and standardization, the development of multiple clustering models, and rigorous evaluation of these models using defined performance metrics. We performed a comparative analysis to identify the most effective clustering technique, followed by labeling the resultant clusters for meaningful insights. This comprehensive process includes a detailed dataset description, results analysis, and a discussion synthesizing our findings from the experimental evaluations.

4.1. Dataset

The dataset utilized in our study, SNS Data, is a collection of Social Networking Services (SNS) data sourced from the widely recognized online platform Kaggle (Skakun, 2020). Kaggle is renowned for its extensive repository of datasets used for data science research and competitions. This dataset encompasses a random sample of 30,000 high school students in the United States profiled on a prominent SNS platform in 2006. To ensure confidentiality, the specific SNS platform remains undisclosed. The dataset represents a diverse student population, with data evenly distributed across four graduating classes from 2006 to 2009. It comprises 40 columns, with the initial four columns providing basic demographic information about each student, including their graduation year, gender, age, and number of friends. The remaining 36 columns are dedicated to various interests expressed by these students on their SNS profiles. These interests are quantified in terms of the frequency of specific words appearing in each student's SNS profile. To facilitate our analysis, we categorized these 36 interest columns into five distinct groups, as outlined in Table 1. This categorization was instrumental in segmenting the dataset for more nuanced analysis and interpretation.

4.2. System requirements

We have tested the proposal on a personal computer. This work can also be tested on *Google Colaboratory*. Test system configuration is provided so that others can compare our results with other machines.

4.3. Tested system

- **Processor:** Intel(R) Core(TM) i9-10900K CPU @ 3.70 GHz
- **Memory:** 32.0 GB RAM
- **Hard Drive:** NVMe Solid State Drive
- **OS:** Windows 11 Pro 64-bit

4.4. Recommended system

- **Processor:** Core i5-7300U 3.5 GHz, AMD Ryzen 3 3300U, or equivalent
- **Memory:** 8 GB RAM or higher
- **Hard Drive:** NVMe Solid State Drive or SATA hard drive
- **OS:** Windows 10 64-bit, Linux-Ubuntu 16.04 or equivalent

4.5. Data preprocessing

After examining the dataset's brief statistics, we need to pre-process the dataset before using it.

- **Treating null-values:** After checking the null value we can see that a total of 5086 records are missing age. And 22054 female, 5222 male adolescent students and 2724 gender missing values. Now we are going to fill all the null values in the gender column with “not disclosed”. The average age, according to the year of graduation:

$$2006 = 19.137241; 2007 = 18.391459; 2008 = 17.523867; 2009 = 16.876025$$

From the summary above, we can notice that the average age varies from roughly one year with every change in the year of graduation. It is helpful to find out if our data is reasonable. We will meet missing values with the average age of each graduating year to deal with these missing values.

- **Convert object variable to numeric:** Since we are going to clustering algorithms we cannot use an objective variable. We need to convert them to numerical variables. So we converted gender into a numerator, 1 for males, 2 for females and 3 for not disclosed.
- **Treating Outliers:** The actual age range is 3–106, which is unrealistic because 3 or 106-year-old students are not teenagers or our target customers. So we can only take a reasonable age limit which is 10–24 for people for our paper.

4.6. Data standardization

A common practice in any analysis using distance calculations is to normalize features or standardize Z-scores so that everyone uses the same range. The Z-score standardization process re-scales properties so that they have an average of zero and a standard deviation of one.

This conversion changes the interpretation of the data to be useful here. In particular, if someone mentions swimming three times in their profile without additional information, we may not realize that they prefer to swim more or less than their peers. If the Z-score is three, we know they mention swimming much more often than the average teenager. After data standardization, we named it `scaled_feature` in our paper.

5. Results analysis

In our study, we rigorously evaluated our proposed methodology using four distinct clustering algorithms. Our first step involved determining the ideal number of clusters for each algorithm, a crucial factor that directly influences the clarity and usefulness of our findings. For this purpose, we created a duplicate of the original dataset. This duplicate was then processed through each of the algorithms to organize the data into clusters. To assess and compare the effectiveness of each algorithm, we utilized the silhouette coefficient as a performance metric. This allowed us to objectively determine which algorithm was most effective, and we subsequently chose the best-performing algorithm to form the basis of our recommended marketing strategies.

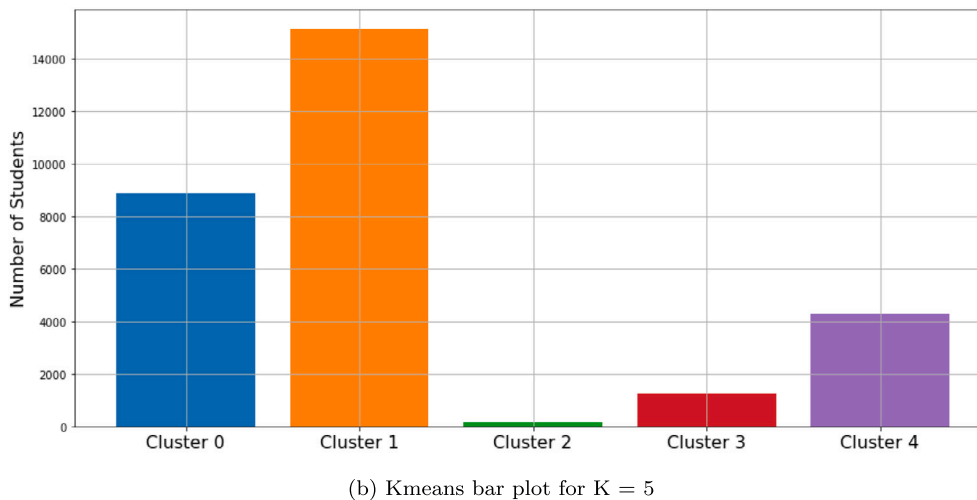
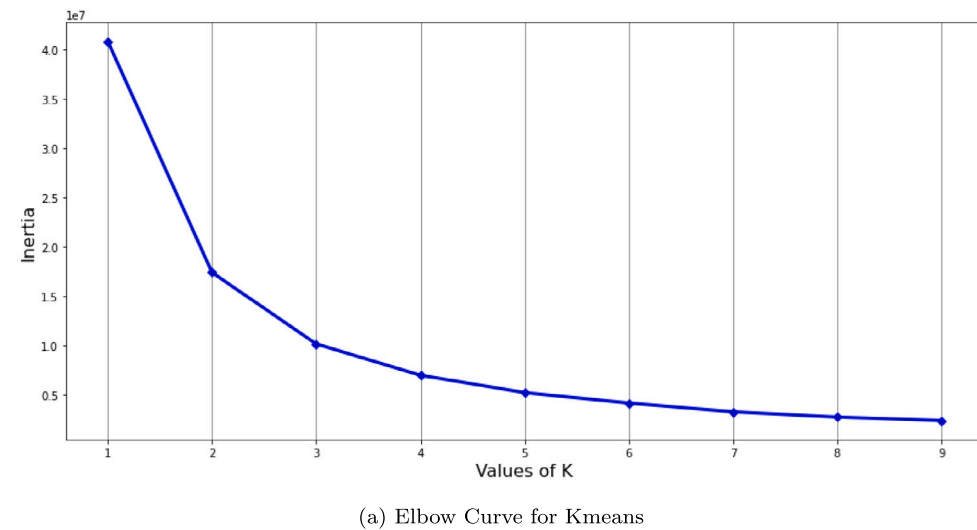


Fig. 3. Kmeans Elbow Curve and Kmeans Result.

5.1. Results of Kmeans clustering

In our study, we employed the K-means clustering algorithm, initially utilizing the elbow method to ascertain the optimal number of clusters. As depicted in Fig. 3a, this method suggests that the ideal number of clusters for our dataset lies between 2 and 3. However, to align with our research objectives, we opted to explore a cluster count of 5. Subsequent assessment of cluster effectiveness was carried out using the silhouette score. This evaluation revealed that the clustering efficiency gradually declined from 2 to 4 clusters, plateauing from 4 to 5 clusters. Nevertheless, we proceeded with 5 clusters for a more detailed segmentation. The final clustering results for K = 5 are illustrated in Fig. 3b, presenting the distribution of data points across the five identified clusters.

5.2. Results of mini-batch K-means clustering

We also utilized Mini-Batch K-means, a variation of the K-means algorithm designed for larger datasets. The silhouette scores for this method, displayed in Fig. 4a, revealed a gradual decline in cluster efficiency from 2 to 4, followed by a noticeable increase in curvature at 5 clusters. Opting for 5 clusters, we observed distinct groupings within the data, as depicted in Fig. 4b. This afforded us a detailed understanding of market segments, crucial for informing our marketing strategy recommendations.

5.3. Results of agglomerative hierarchical clustering

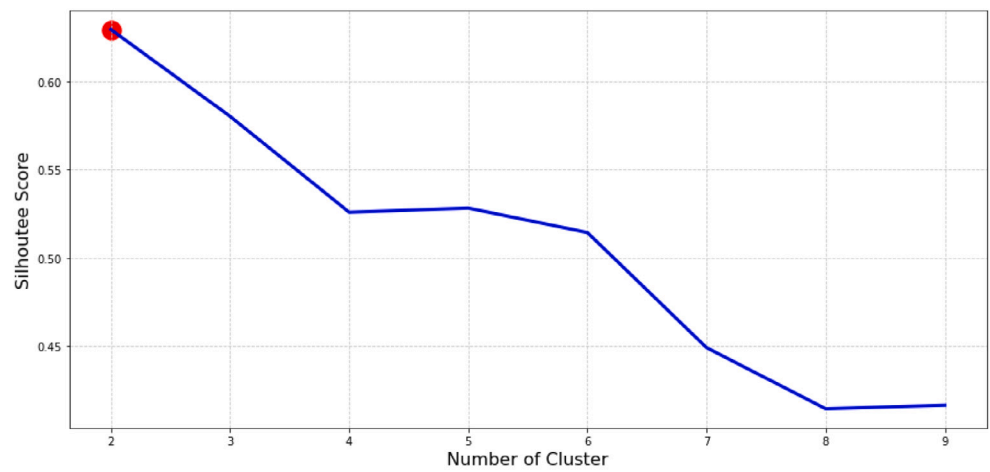
To explore hierarchical structures, we employed Agglomerative Hierarchical Clustering, a bottom-up approach. The dendrogram in Fig. 5 revealed natural groupings in the dataset, leading us to choose 5 clusters.

The silhouette analysis, as seen in Fig. 6a, indicated diminishing returns in clustering efficiency for AGNES. However, our choice to utilize 5 clusters aimed at achieving a more detailed segmentation. Fig. 6b displays the clustering outcome for 5 clusters, aiding our understanding of the diverse customer groups within our dataset.

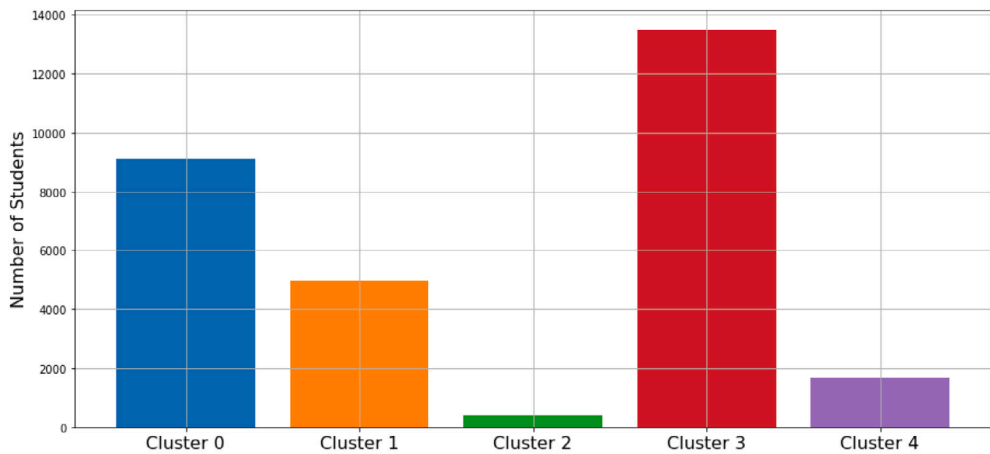
5.4. Results of Fuzzy C-means clustering

The results of Table 2 help us to verify the clustering. We know that the higher the value of PC [Partition Coefficient], the better the result of the partition. And the lower the value of PE [Partition Entropy], the better the result of the partition.

We tested the effectiveness of our clusters for Fuzzy C-means clustering. This is shown by the silhouette score graph in Fig. 7a which measures the performance of our clusters for Fuzzy C-means. We see that from 2 to 9 the efficiency of the cluster decreases almost linearly. So let us see the result of C-means in Fig. 7b where the number of clusters is 5.



(a) Silhouette curve to predict the best clustering performance of MBK



(b) Mini-batch K-means bar plot for K = 5

Fig. 4. Mini-batch Silhouette curve and K-means.

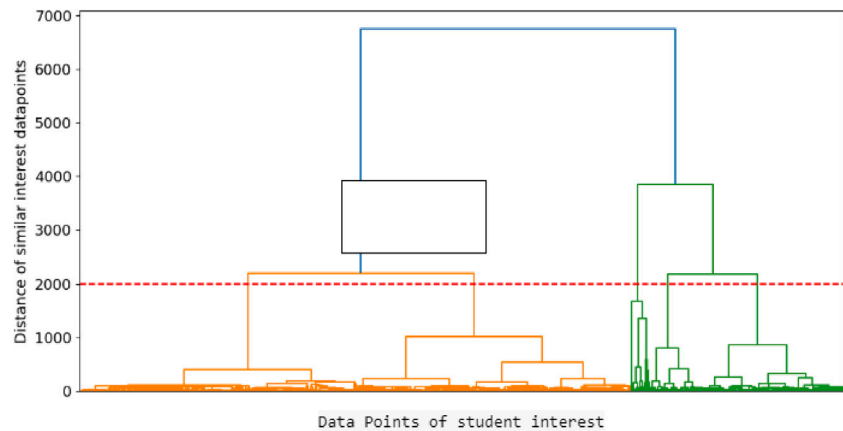
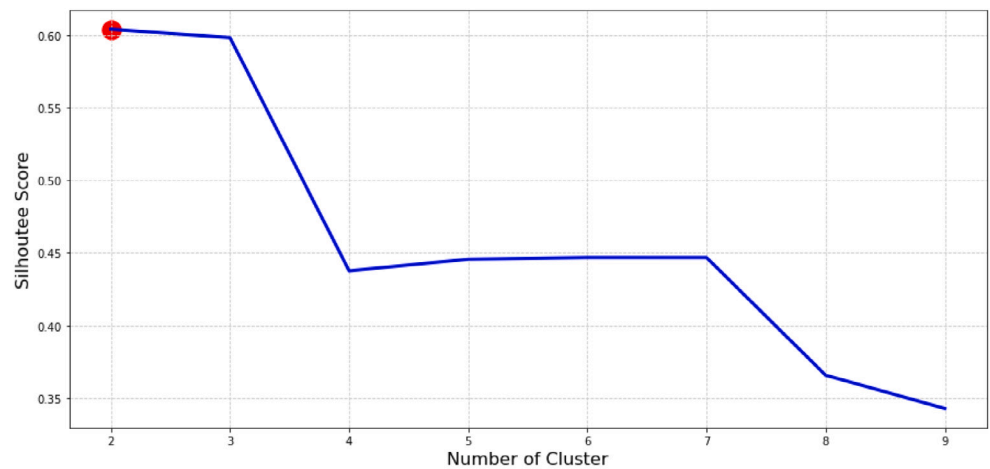
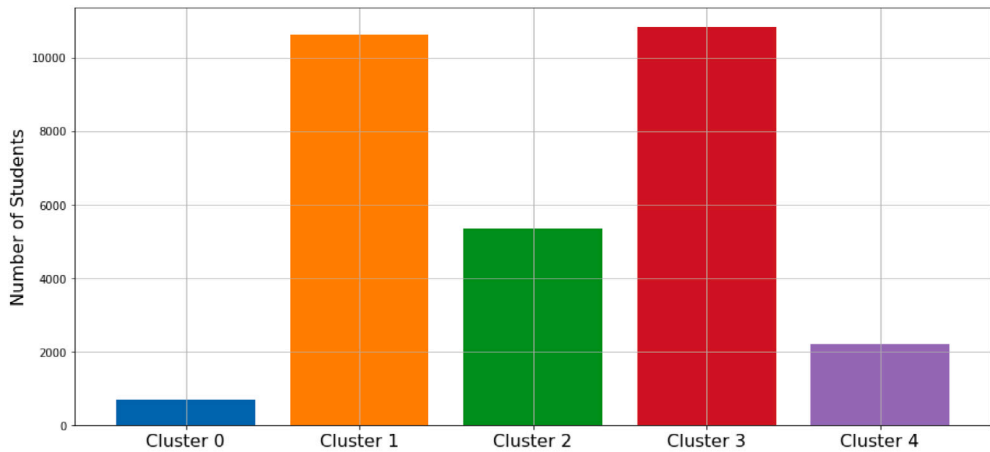


Fig. 5. Dendrogram to show hierarchical relationships.



(a) Silhouette curve to predict the best clustering performance of AGNES



(b) Agglomerative hierarchical clustering bar plot for K = 5

Fig. 6. Combined figure for AGNES with subfloats.

Table 2
PC and PE values to predict the best number of clusters for C-means.

Cluster	Partition Coefficient [PC]	Partition Entropy [PE]
2	0.4298	0.1727
3	0.2665	0.1757
4	0.1887	0.1676
5	0.1433	0.1596
6	0.1139	0.1514
7	0.0931	0.1451
8	0.0766	0.1432
9	0.0640	0.1413

Each of these clustering methods provided unique insights into our data, helping us to comprehensively understand and segment the digital native market for strategic marketing applications.

5.5. Efficiency verification

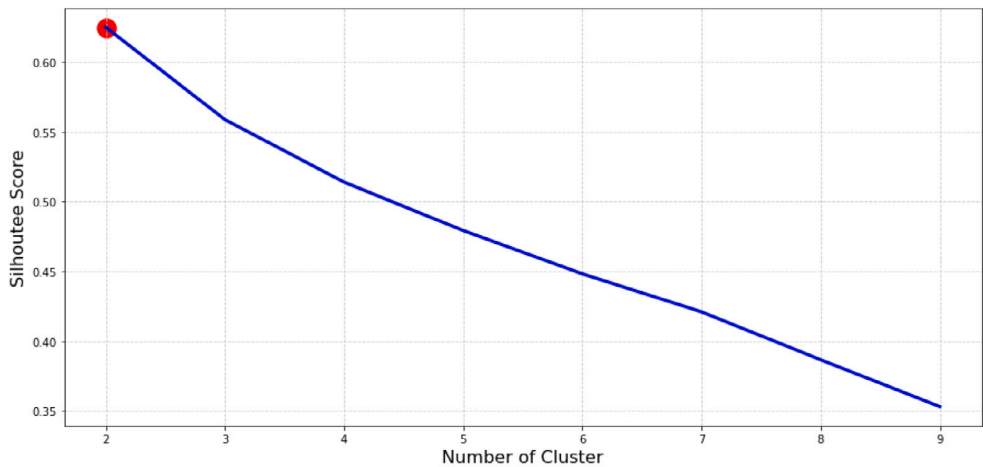
Our objective is to categorize the interests of teenagers into five groups. To achieve the best outcome, we opted for five clusters in total. We used the Silhouette Score to assess the clustering performance. After assessing the four clustering algorithms for a cluster number of five, we determined that the Mini-batch K-means algorithm produced the highest silhouette score of 0.52820. Fig. 8 presents a comparison of the clustering performance.

5.6. Discussion

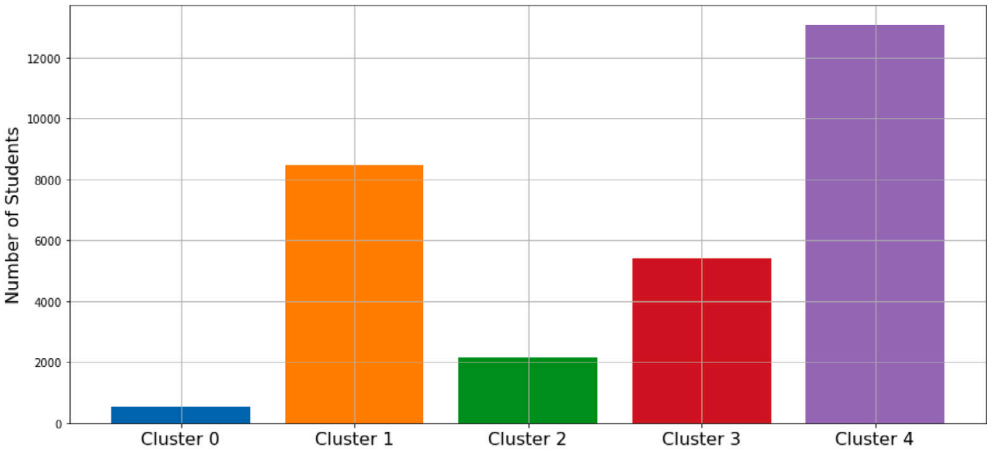
Our study aimed to discern distinct groups among adolescent users based on their shared interests. This segmentation offers invaluable insights for marketing professionals, enabling targeted and effective marketing strategies both in the short and long term. For instance, consider a company specializing in sporting goods. By leveraging our segmented results, they can tailor their product offerings to different clusters, optimizing their advertising and promotional efforts accordingly. This approach facilitates the collection of targeted marketing feedback, enabling iterative improvements in product positioning and messaging.

To facilitate further analysis and application, we organized the diverse range of interests into five distinct categories, as illustrated in Table 1. This categorization process enabled the creation of a streamlined dataset, enhancing its usability and interpretability. The resulting dataset, depicted in Table 3, provides a clear overview of the structured data format.

Fig. 9a presents the distribution of students across clusters based on their demonstrated interest in specific sections, indicating the percentage of students within each cluster who have expressed interest in a given section at least once. Conversely, students not represented in the chart did not exhibit interest in the corresponding section. Complementing this, Fig. 9b depicts the average scores of each cluster across five distinct categories.



(a) Silhouette curve to predict the best clustering performance of Fuzzy C-means



(b) Fuzzy C-means clustering bar plot for K = 5

Fig. 7. Silhouette curve and Fuzzy C-means clustering.

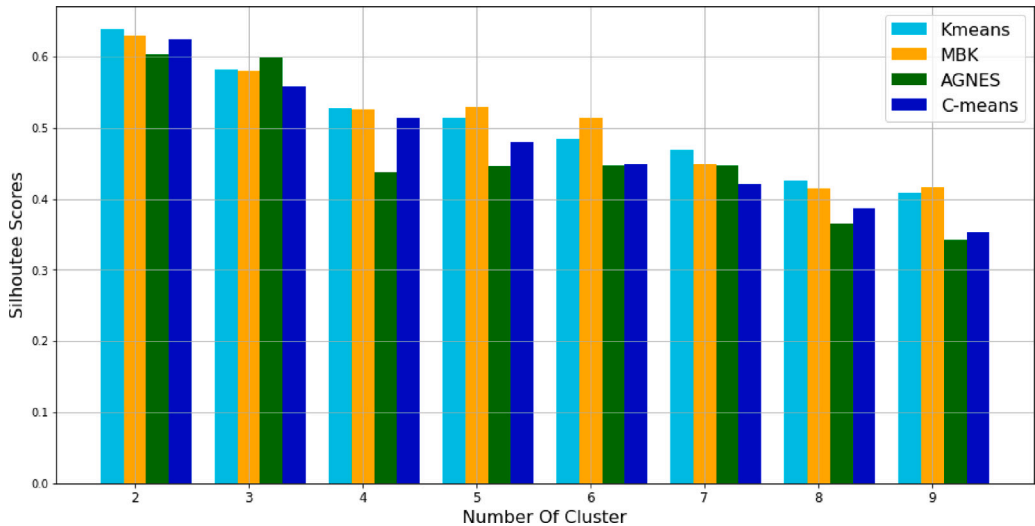


Fig. 8. Comparison of clustering performance on SNS data.

Table 3
New SNS dataset head.

cluster	grad year	gender	age	friends	Athletes	Extra_activities	Fashion	Religious	Antisocial
0	2	2006	1	18.9820	7	0	1	0	0
1	2	2006	2	18.8010	0	1	5	11	0
2	3	2006	1	18.3350	69	1	3	0	1
3	2	2006	2	18.8750	0	0	2	0	0
4	2	2006	3	18.9950	10	0	5	3	8
5	1	2006	2	19.1372	142	0	3	2	2

Table 4
Comparison analysis of clustering techniques on SNS data.

Cluster numbers	Kmeans	MiniBatch Kmeans	AGNES	Fuzzy C-means
2	0.6391	0.6295	0.6038	0.6246
3	0.5807	0.5803	0.5980	0.5586
4	0.5269	0.5258	0.4376	0.5138
5	0.5135	0.5282	0.4456	0.4793
6	0.4847	0.5143	0.4467	0.4482
7	0.4687	0.4489	0.4467	0.4209
8	0.4261	0.4142	0.3658	0.3865
9	0.4081	0.4163	0.3429	0.3531

Upon examining these statistical insights, suppose an individual seeks to promote their business within the sports sector. In such a scenario, a prudent approach would involve evaluating the ‘Athlete’ category. Notably, Fig. 9a reveals that the preponderance of individuals interested in athletics primarily resides within Cluster 2, followed by Clusters 0 and 3. Further analysis, as depicted in Fig. 9b, highlights that Cluster 3 exhibits the highest average score for the ‘Athlete’ category, followed closely by Cluster 2.

With these valuable insights, marketing experts can make informed decisions to optimize their promotional strategies. For instance, in scenarios where budgetary constraints dictate a selective approach, targeting Cluster 3 may prove advantageous due to its favorable engagement metrics. Conversely, if budgetary considerations allow for broader targeting, prioritizing Cluster 2, with its sizable ‘Athlete’ population, could yield enhanced promotional efficacy.

Ultimately, the selection of target consumers rests within the purview of marketing professionals, who can leverage the analytical framework provided by our study to tailor their strategies to the nuanced preferences and behaviors of distinct consumer clusters. By integrating these findings into their marketing endeavors, practitioners can optimize resource allocation and maximize the impact of their promotional efforts.

5.7. Comparison analysis

This section presents a comparative assessment of four distinct clustering methodologies: K-means, MiniBatch K-means, AGNES (Agglomerative Hierarchical Clustering), and Fuzzy C-means. The evaluation, outlined in Table 4, is based on silhouette scores computed across varying cluster numbers spanning from 2 to 9. Silhouette score, a metric gauging the similarity of an object to its cluster relative to others, offers insights into the efficacy of clustering algorithms, with higher scores indicative of more clearly defined clusters.

The results in Table 4 underscore several notable observations. Notably, K-means consistently exhibits superior silhouette scores across diverse cluster configurations, indicating its efficacy in generating distinct and well-separated clusters. Conversely, MiniBatch K-means, a variant of the traditional K-means algorithm designed for handling large datasets, demonstrates comparable performance but exhibits nuanced efficiency fluctuations, particularly evident in configurations involving five and six clusters.

In contrast, AGNES showcases diminishing silhouette scores with increasing cluster numbers, suggestive of challenges in delineating distinct clusters at higher cluster counts. This trend aligns with the hierarchical nature of AGNES, wherein clusters are incrementally merged,

potentially leading to diminished cluster separability at higher levels of aggregation.

Furthermore, Fuzzy C-means displays a gradual decline in performance as the number of clusters escalates, mirroring its inherent characteristic of soft clustering, where data points may possess varying degrees of membership across multiple clusters.

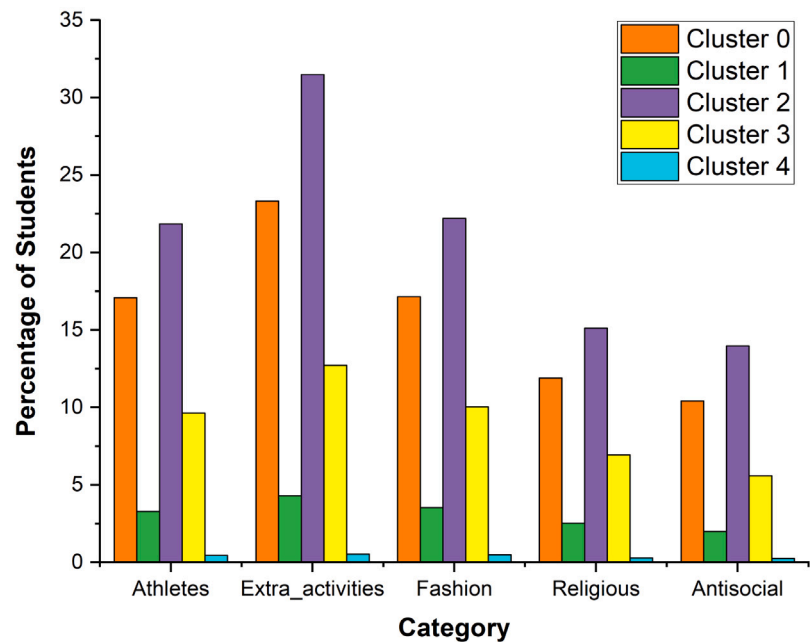
In summary, while each clustering technique exhibits distinct strengths and limitations, K-means and MiniBatch K-means emerge as preferred choices owing to their consistent and elevated silhouette scores, especially for scenarios involving a lower number of clusters. However, the selection of an appropriate clustering methodology must be contingent upon the specific characteristics of the dataset under consideration and the desired granularity of market segmentation objectives.

5.8. Fruitful clustering business or profit recommendation of SNS data

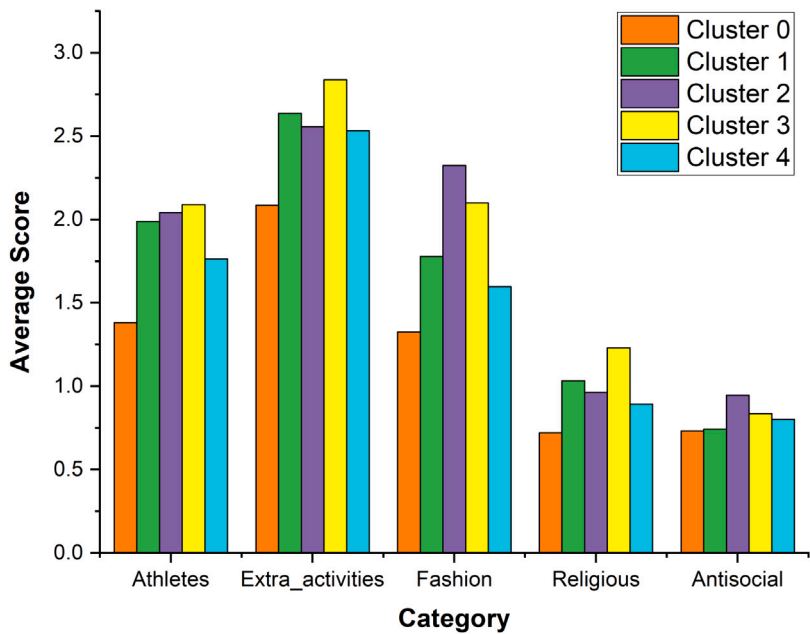
The utilization of clustering techniques on Social Networking Services (SNS) data opens up numerous avenues for businesses to enhance their strategies. Our findings, derived from the meticulous clustering of SNS data, provide valuable insights that can be leveraged in various business contexts.

1. Targeted Marketing Strategies One of the most direct applications of our clustering findings is in the realm of targeted marketing. By understanding the distinct characteristics and preferences of each cluster, businesses can tailor their marketing campaigns to resonate more effectively with each group. This customization can lead to higher engagement rates, improved customer satisfaction, and increased return on investment in marketing efforts.
2. Product Development and Innovation Our clustering results offer a rich source of information for product development. Companies can identify unmet needs and preferences within each cluster, guiding the innovation of new products or the enhancement of existing ones. This data-driven approach ensures that product development is aligned with actual consumer demands, thereby increasing the likelihood of market success.
3. Customer Engagement and Loyalty Programs By understanding the behavior and preferences of different clusters, businesses can design customer engagement and loyalty programs that are more appealing to each group. Personalized offers, rewards, and communications based on clustering insights can lead to deeper customer engagement and long-term loyalty.
4. Strategic Business Decisions The insights from our clustering analysis can also inform broader strategic decisions. For instance, identifying a cluster with a high potential for growth can lead a business to allocate more resources to target this group. Similarly, understanding the evolving trends within each cluster can assist in long-term planning and forecasting.

Incorporating our clustering findings into business strategies offers a wealth of opportunities for enhanced performance, customer satisfaction, and profitability. The potential applications outlined in this section are just a starting point, and we encourage businesses to explore further ways to utilize these insights for their unique contexts and objectives.



(a) The Percentage of students in each cluster who have shown their interest at least once.



(b) Average score of each cluster for the five categories.

Fig. 9. Comparison of interest percentages and average category scores for each cluster.

5.9. Implications of our proposal

Our paper indeed addresses several implications for research, practice, and society. Here’s how we bridge the gap between theory and practice:

1. Research Implications: Our study contributes to the body of knowledge by introducing a novel Clustering-Based Market Analysis Model. Researchers can further explore and extend our model to investigate other aspects of market analysis, leading to advancements in the field.

2. Practical Applications: Practitioners in marketing and business can leverage our model to enhance their market analysis efforts. By segmenting customer groups based on shared interests, businesses can tailor their marketing strategies more effectively, resulting in improved sales and customer satisfaction.
3. Teaching and Training: Our research findings can be incorporated into academic curricula and training programs for marketing and data analysis. By teaching students about advanced clustering techniques and their applications in market analysis, educators can prepare future professionals for the evolving landscape of marketing.

4. Public Policy and Society: The insights derived from our research can inform public policies aimed at promoting sustainable and healthy city environments. By leveraging social media data for early prediction of COVID-19 outbreaks, policymakers can implement proactive measures to mitigate the spread of the virus and safeguard public health.

In summary, our paper's implications are consistent with the findings and conclusions presented. We believe that our research has the potential to make meaningful contributions to both academia and real-world practice, ultimately benefiting society at large.

6. Conclusion

In this paper, we have demonstrated the significant role of customer segmentation in enhancing the understanding of business organizations about their customers, especially digital natives. The surge of startups and businesses focused on this demographic highlights the importance of innovative marketing strategies. In this context, social networking sites emerge as a vital source of data, offering rich insights into customer interests and behaviors.

Our contribution lies in the development and implementation of a clustering approach that leverages social networking site (SNS) data to segment digital native customers effectively. By utilizing a prototype dataset for testing, we have successfully extracted and analyzed valuable information from SNS data. This involved a meticulous process of labeling and categorizing the data into five distinct groups based on 36 varied interests. The strength of our approach is evidenced by the comprehensive testing and evaluation of our model, which demonstrates its capability to differentiate digital natives into various groups based on their specific interests. The implications of our findings are far-reaching for marketing strategists. The insights derived from our model offer a roadmap for tailoring advertising campaigns more effectively and developing marketing strategies that resonate with the dynamic preferences of digital natives.

In conclusion, our research provides a novel framework for businesses to understand and engage with a crucial customer segment. The methodology and findings of this study have the potential to transform marketing approaches, making them more data-driven and aligned with the evolving landscape of digital consumer behavior.

7. Limitation and future work

We have acknowledged the key constraints of our study. Firstly, we discuss how the dependency of our clustering model on the quality and comprehensiveness of market data may not fully encapsulate the diverse range of user interests. Lastly, we highlight the challenge of keeping up with rapid changes in market trends and consumer behaviors, which could affect the model's relevance over time.

Our future work aims to improve our model's adaptability to diverse market environments by incorporating a wider array of data sources, including real-time market trends, to enhance accuracy and relevance. We plan to collaborate with business organizations to test and refine our model, ensuring it aligns with industry-specific needs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was partially supported by CREST at the Deakin Cyber Research and Innovation Centre (DEAKIN CYBER), Australia.

References

- Allaway, A. W., McCullough, H., Kim, K., & McAmis, G. (2022). Segmenting customer adoption of new consumer packaged goods using latent growth curve modeling. *Journal of Marketing Theory and Practice*, 1–18.
- Almomani, A., Nahar, K., Alauthman, M., Al-Betar, M. A., Yaseen, Q., & Gupta, B. B. (2024). Image cyberbullying detection and recognition using transfer deep machine learning. *International Journal of Cognitive Computing in Engineering*, 5, 14–26.
- Andrews, L., Bianchi, C., Wiese, M., Cuneo, A., & Fazal E. Hasan, S. (2019). Segmenting brands' social network site (Sns) consumers: a four-country study. *Journal of International Consumer Marketing*, 31(1), 22–38.
- Ashari, I., Nugroho, E., Baraku, R., Yanda, I., & Liwardana, R. (2023). Analysis of Elbow, Silhouette, Davies-Bouldin, Calinski-Harabasz, and Rand-Index evaluation on K-means algorithm for classifying flood-affected areas in Jakarta. *Journal of Applied Informatics and Computing*, 7(1), 95–103.
- Azzaoui, A. E., Singh, S. K., & Park, J. H. (2021). SNS big data analysis framework for COVID-19 outbreak prediction in smart healthy city. *Sustainable Cities and Society*, 71, Article 102993.
- Balasubramani, S., Aravindhar, J., Renjith, P., & Ramesh, K. (2024). DDSS: Driver decision support system based on the driver behaviour prediction to avoid accidents in intelligent transport system. *International Journal of Cognitive Computing in Engineering*, 5, 1–13.
- Bhattacharjya, A., Islam, M. M., Uddin, M. A., Talukder, M. A., Azad, A., Aryal, S., et al. (2024). Exploring Gene Regulatory Interaction Networks and predicting therapeutic molecules for Hypopharyngeal Cancer and EGFR-mutated lung adenocarcinoma. arXiv preprint arXiv:2402.17807.
- Bottaro, R., & Faraci, P. (2022). The use of social networking sites and its impact on adolescents' emotional well-being: a scoping review. *Current Addiction Reports*, 9(4), 518–539.
- Cartwright, S., & Davies, I. A. (2022). The development of B2B social networking capabilities. *Industrial Marketing Management*, 106, 139–151.
- Choi, J. Y., Jung, G., Kim, Y., & Kim, Y. B. (2020). Simulation analysis of multi-group competitive relationships between platforms in social network service (SNS) market. *Journal of the Korea Society for Simulation*, 29(4), 9–19.
- de Almeida, G. S., Rizzo, R., Amorim, M. T. A., dos Santos, N. V., Rosas, J. T. F., Campos, L. R., et al. (2023). Monitoring soil-plant interactions and maize yield by satellite vegetation indexes, soil electrical conductivity and management zones. *Precision Agriculture*, 1–21.
- Demircan, M., Seller, A., Abut, F., & Akay, M. F. (2021). Developing Turkish sentiment analysis models using machine learning and e-commerce data. *International Journal of Cognitive Computing in Engineering*, 2, 202–207.
- Ezugwu, A. E., Ikotun, A. M., Oyelade, O. O., Abualigah, L., Agushaka, J. O., Eke, C. I., et al. (2022). A comprehensive survey of clustering algorithms: State-of-the-art machine learning applications, taxonomy, challenges, and future research prospects. *Engineering Applications of Artificial Intelligence*, 110, Article 104743.
- Hanji, S., & Hanji, S. (2023). Towards performance overview of mini batch K-means and K-means: Case of four-wheeler market segmentation. In *International conference on smart trends in computing and communications* (pp. 801–813). Springer.
- Hasan, M. R., Alazab, A., Joy, S. B., Uddin, M. N., Uddin, M. A., Khraisat, A., et al. (2023). Smart contract-based access control framework for internet of things devices. *Computers*, 12(11), 240.
- Hassan, A. U., Hussain, J., Hussain, M., Sadiq, M., & Lee, S. (2017). Sentiment analysis of social networking sites (SNS) data using machine learning approach for the measurement of depression. In *2017 international conference on information and communication technology convergence* (pp. 138–140). IEEE.
- Ikotun, A. M., Ezugwu, A. E., Abualigah, L., Abuhajja, B., & Heming, J. (2023). K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data. *Information Sciences*, 622, 178–210.
- Indartoyo, I. (2021). Consumer selection of e-marketplace type based on demographic and behavioral segmentation. *PalArch's Journal of Archaeology of Egypt/Egyptology*, 18(1), 322–332.
- Islam, M. M., Adil, M. A. A., Talukder, M. A., Ahamed, M. K. U., Uddin, M. A., Hasan, M. K., et al. (2023). DeepCrop: Deep learning-based crop disease prediction with web application. *Journal of Agriculture and Food Research*, 14, Article 100764.
- Islam, M. M., Talukder, M. A., Sarker, M. R. A., Uddin, M. A., Akhter, A., Sharmin, S., et al. (2023). A deep learning model for cotton disease prediction using fine-tuning with smart web application in agriculture. *Intelligent Systems with Applications*, 20, Article 200278.
- Kaur, S., Singh, S., & Kaushal, S. (2024). Deep learning-based approaches for abusive content detection and classification for multi-class online user-generated data. *International Journal of Cognitive Computing in Engineering*.
- Khatun, M., Islam, M. M., Rifat, H. R., Shahid, M. S. B., Talukder, M. A., & Uddin, M. A. (2023). Hybridized convolutional neural networks and long short-term memory for improved Alzheimer's disease diagnosis from MRI scans. In *2023 26th international conference on computer and information technology* (pp. 1–6). IEEE.
- Kolli, S., Salavatian, S., Khajeheian, D., & Yazdani, A. Identifying Generation Z's Behavioral Patterns in Social Media: A Case Study of Big Data Generated from Active Teens in Instagram.

- Krishnamoorthy, P., Sathiyarayanan, M., & Proença, H. P. (2024). A novel and secured email classification and emotion detection using hybrid deep neural network. *International Journal of Cognitive Computing in Engineering*, 5, 44–57.
- Lorenzo-Romero, C., Constantinides, E., & Alarcón-del Amo, M.-d.-C. (2012). Segmenting the social networking sites users: an empirical study. *International Journal of Internet Marketing and Advertising*, 7(2), 136–156.
- Manor, I., & Kampf, R. (2022). Digital nativity and digital diplomacy: Exploring conceptual differences between digital natives and digital immigrants. *Global Policy*, 13(4), 442–457.
- Marc, P. (2001). Digital natives, digital immigrants. *On the Horizon*, 9(5), 1–6.
- Mazzocchi, F. J., & Lucarelli, C. (2023). Success or failure in equity crowdfunding? A systematic literature review and research perspectives. *Management Research Review*, 46(6), 790–831.
- Naim, A. (2023). Consumer behavior in marketing patterns, types, segmentation. *European Journal of Economics, Finance and Business Development*, 1(1), 1–18.
- Nazari, M., Hussain, A., & Musilek, P. (2023). Applications of clustering methods for different aspects of electric vehicles. *Electronics*, 12(4), 790.
- Nikou, S., Mezei, J., & Brännback, M. (2018). Digital natives' intention to interact with social media: Value systems and gender. *Telematics and Informatics*, 35(2), 421–435.
- Patel, P., Sivaiah, B., & Patel, R. (2022). Approaches for finding optimal number of clusters using k-means and agglomerative hierarchical clustering techniques. In *2022 international conference on intelligent controller and computing for smart power* (pp. 1–6). IEEE.
- Peters, A., Rohr, L., & Squires, L. (2022). Examining social media in the online classroom: postsecondary students' Twitter use and motivations. *International Journal of Social Media and Interactive Learning Environments*, 6(4), 328–348.
- Qadadeh, W., & Abdallah, S. (2018). Customers segmentation in the insurance company (TIC) dataset. *Procedia Computer Science*, 144, 277–290.
- Quraishi, M. I., & Choudhury, J. P. (2023). Assessment, categorisation and prediction of the landslide-affected regions using soft computing and clustering techniques. *Journal of the Institution of Engineers (India): Series B*, 1–24.
- Rad, D., Cuc, L. D., Lile, R., Balas, V. E., Barna, C., Pantea, M. F., et al. (2022). A cognitive systems engineering approach using unsupervised fuzzy C-means technique, exploratory factor analysis and network analysis—A preliminary statistical investigation of the bean counter profiling scale robustness. *International Journal of Environmental Research and Public Health*, 19(19), 12821.
- Rana, M. M., Islam, M. M., Talukder, M. A., Uddin, M. A., Aryal, S., Alotaibi, N., et al. (2023). A robust and clinically applicable deep learning model for early detection of Alzheimer's. *IET Image Processing*.
- Ranto, D. W. P., Hariningsih, E., Prasetyanto, W. E., & Oktafiani, D. M. (2021). The price discounts and fashion involvement to increase online impulsive buying: Study among teenagers at Yogyakarta. *International Journal of Business, Management and Economics*, 2(4), 239–250.
- Sharmin, S., Ahammad, T., Talukder, M. A., & Ghose, P. (2023). A hybrid dependable deep feature extraction and ensemble-based machine learning approach for breast cancer detection. *IEEE Access*.
- Skakun, T. (2020). Social network services data. Retrieved from <https://www.kaggle.com/datasets/artemskakun/snsdata>. (Accessed 19 December 2022).
- Smith, K. T. (2019). Mobile advertising to Digital Natives: preferences on content, style, personalization, and functionality. *Journal of Strategic Marketing*, 27(1), 67–80.
- Talukder, M. A., Hasan, K. F., Islam, M. M., Uddin, M. A., Akhter, A., Yousuf, M. A., et al. (2023). A dependable hybrid machine learning model for network intrusion detection. *Journal of Information Security and Applications*, 72, Article 103405.
- Talukder, M. A., Hossen, R., Uddin, M. A., Uddin, M. N., & Acharjee, U. K. (2024). Securing transactions: A hybrid dependable ensemble machine learning model using IHT-LR and grid search. *Cybersecurity*, arXiv:2402.14389.
- Talukder, M. A., Islam, M. M., Uddin, M. A., Akhter, A., Hasan, K. F., & Moni, M. A. (2022). Machine learning-based lung and colon cancer detection using deep feature extraction and ensemble learning. *Expert Systems with Applications*, 205, Article 117695.
- Talukder, M. A., Islam, M. M., Uddin, M. A., Akhter, A., Pramanik, M. A. J., Aryal, S., et al. (2023). An efficient deep learning model to categorize brain tumor using reconstruction and fine-tuning. *Expert Systems with Applications*, Article 120534.
- Talukder, M. A., Islam, M. M., Uddin, M. A., Hasan, K. F., Sharmin, S., Alyami, S. A., et al. (2024). Machine learning-based network intrusion detection for big and imbalanced data using oversampling, stacking feature embedding and feature extraction. *Journal of Big Data*, 11(1), 1–44.
- Talukder, M. A., Layek, M. A., Kazi, M., Uddin, M. A., & Aryal, S. (2023). Empowering COVID-19 detection: Optimizing performance through fine-tuned EfficientNet deep learning architecture. *Computers in Biology and Medicine*, Article 107789.
- Talukder, M. A., Sharmin, S., Uddin, M. A., Islam, M. M., & Aryal, S. (2024). MLSTL-WSN: machine learning-based intrusion detection using SMOTETomek in WSNs. *International Journal of Information Security*, 1–20.
- Thimm, C. (2023). Mediatized families: Digital parenting on social media. In *Families and new media: Comparative perspectives on digital transformations in law and society* (pp. 33–57). Springer Fachmedien Wiesbaden Wiesbaden.
- Umuhoza, E., Ntirushwamaboko, D., Awuah, J., & Birir, B. (2020). Using unsupervised machine learning techniques for behavioral-based credit card users segmentation in Africa. *SAIEE Africa Research Journal*, 111(3), 95–101.
- Wang, C. (2022). Efficient customer segmentation in digital marketing using deep learning with swarm intelligence approach. *Information Processing & Management*, 59(6), Article 103085.
- Wang, H.-Y., Wang, J.-S., & Wang, G. (2022). A survey of fuzzy clustering validity evaluation methods. *Information Sciences*.
- Wong, L.-W., Tan, G. W.-H., Hew, J.-J., Ooi, K.-B., & Leong, L.-Y. (2020). Mobile social media marketing: a new marketing channel among digital natives in higher education? *Journal of Marketing for Higher Education*, 1–25.
- Wongkhuenkaew, R., Auephanwiriyakul, S., Theera-Umpon, N., Teeyapan, K., & Yeesarapat, U. (2023). Fuzzy K-nearest neighbor based dental fluorosis classification using multi-prototype unsupervised possibilistic fuzzy clustering via cuckoo search algorithm. *International Journal of Environmental Research and Public Health*, 20(4), 3394.
- Zhao, C., Wu, M., Liu, J., Duan, Z., Shen, L., Shangguan, X., et al. (2023). Progress and prospects of data-driven stock price forecasting research. *International Journal of Cognitive Computing in Engineering*, 4, 100–108.